

ARDL Summary from RW4 Primary Chain

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

RW4 domestic vertical chain is FarmGateUA -> ProducerUA -> ProZorro -> Retail, estimated with both forward and reverse-flow pairs. Farm-gate enters from two alternative reconstruction workbooks and both linear/pchip interpolation variants are carried end-to-end. Retail_combined is the anchored downstream index built from Silpo effective prices, Novus observed prices, and a level-aligned ConsumerUA anchor, while Retail_combined_core keeps the strict retailer-only overlap for robustness.

family=ARDL

rows=1383

source=/Users/getapple/Documents/KSE/Master Thesis/Main materials/Model/Charniuk_Dairy_Research/outputs/primary_chain_summary/primary_chain_consolidated.xlsx

Table: ARDL_Summary

panel_level	panel_name	segment_name	product	standardized_type	retailer_panel	brand	price_variant	reconstruction_variant	farm_gate	source	frequency	intersection_rule	pair_scope	chain_direction	stage_from	stage_to	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	unreliable_reason	shock_dummy_included	promo_controls_included	ljungbox_p	arch_p	jb_p	notes	robust_linear_vs_pchip	interpolation_sensitive	robust_across_reconstruction	core_finding_flag
pairwise_product	pairwise::farm_gate_produ...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	forward	FarmGateUA	ProducerUA	farmgateua_to_producerua	ARDL	ProducerUA	FarmGateUA	1460	1.008292630698435	1.148133997300111	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	3.394435607307315e-48	2.657620106201015e-36	0.0	bounds_lower_p=6.854e-07;...	1	0	0		
pairwise_product	pairwise::farm_gate_produ...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	reverse	ProducerUA	FarmGateUA	producerua_to_farmgateua	ARDL	FarmGateUA	ProducerUA	1460	0.08354380077931722	0.7466498335912217	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.0001667520252207402	1.487068813072334e-33	0.0	bounds_lower_p=0.008057;...	1	0	0		
pairwise_product	pairwise::farm_gate_produ...	Кефір	Кефір	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	forward	FarmGateUA	ProducerUA	farmgateua_to_producerua	ARDL	ProducerUA	FarmGateUA	1460	0.8457378389145287	0.718354700672924	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	1.408996758184214e-48	2.041030793652789e-41	0.0	bounds_lower_p=6.087e-08;...	1	0	0		
pairwise_product	pairwise::farm_gate_produ...	Кефір	Кефір	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	reverse	ProducerUA	FarmGateUA	producerua_to_farmgateua	ARDL	FarmGateUA	ProducerUA	1460	0.05822192871835696	1.037038749734055	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	2.139201233577975e-05	2.888195495062957e-37	0.0	bounds_lower_p=0.1237; bo...	1	0	0		
pairwise_product	pairwise::farm_gate_produ...	Молоко питне	Молоко питне	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	forward	FarmGateUA	ProducerUA	farmgateua_to_producerua	ARDL	ProducerUA	FarmGateUA	1460	0.5773740475787523	0.2366729545920248	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	1.609546133018051e-50	2.848808722494858e-40	0.0	bounds_lower_p=9.83e-08;...	1	0	0		
pairwise_product	pairwise::farm_gate_produ...	Молоко питне	Молоко питне	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	reverse	ProducerUA	FarmGateUA	producerua_to_farmgateua	ARDL	FarmGateUA	ProducerUA	1460	0.05148394703596271	1.651470907241549	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	2.46781234667315e-06	3.536484520773019e-39	0.0	bounds_lower_p=0.5295; bo...	1	0	0		
pairwise_product	pairwise::farm_gate_produ...	Сир твердий	Сир твердий	hard_cheese	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	forward	FarmGateUA	ProducerUA	farmgateua_to_producerua	ARDL	ProducerUA	FarmGateUA	1460	0.8828799479012925	0.6040667068810439	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	1.788308342964432e-52	1.135545133629706e-39	0.0	bounds_lower_p=6.9e-08; b...	1	0	0		
pairwise_product	pairwise::farm_gate_produ...	Сир твердий	Сир твердий	hard_cheese	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	reverse	ProducerUA	FarmGateUA	producerua_to_farmgateua	ARDL	FarmGateUA	ProducerUA	1460	0.0681508540616691	1.164422143448012	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	1.686952992080812e-05	1.465431504934138e-34	0.0	bounds_lower_p=0.2163; bo...	1	0	0		
pairwise_product	pairwise::farm_gate_produ...	Сметана	Сметана	sour_cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	forward	FarmGateUA	ProducerUA	farmgateua_to_producerua	ARDL	ProducerUA	FarmGateUA	1460	0.8985306219908011	0.7596196625063331	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	3.554835035137000e-47	2.784121796816722e-34	0.0	bounds_lower_p=7.586e-08;...	1	0	0		
pairwise_product	pairwise::farm_gate_produ...	Сметана	Сметана	sour_cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	reverse	ProducerUA	FarmGateUA	producerua_to_farmgateua	ARDL	FarmGateUA	ProducerUA	1460	0.06828470814816878	0.9864247198873515	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	6.662109560315183e-05	1.385144966609479e-35	0.0	bounds_lower_p=0.1749; bo...	1	0	0		
pairwise_product	pairwise::farm_gate_prozo...	Сир твердий	Сир твердий	hard_cheese	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_prozorro	forward	FarmGateUA	ProZorro	farmgateua_to_prozorro	ARDL	ProZorro	FarmGateUA	191	-7.784607141579695	-0.6602991663084279	nan	nan	nan	nan	ok	0	0	0	0.4527885251852568	0.8163134284212645	0.0001313557362105091	0.0	0	0	0	0			
pairwise_product	pairwise::producer_prozor...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	forward	FarmGateUA	ProducerUA	farmgateua_to_producerua	ARDL	ProducerUA	FarmGateUA	220	-0.3414020826429578	2.412553156998073	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.8084456638086809	7.707421694704969e-06	0.0	bounds_lower_p=0.05771; b...	1	0	0		
pairwise_product	pairwise::producer_prozor...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	reverse	ProducerUA	FarmGateUA	producerua_to_farmgateua	ARDL	FarmGateUA	ProducerUA	220	-0.009496304094984254	-0.01746214807732793	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.9668396324319617	0.0009199039352391156	1.483129002978034e-17	bounds_lower_p=4.974e-07;...	1	0	0		
pairwise_product	pairwise::producer_prozor...	Молоко питне	Молоко питне	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	forward	FarmGateUA	ProducerUA	farmgateua_to_producerua	ARDL	ProducerUA	FarmGateUA	216	-1.114357960499774	2.518467879222956	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.4538007660294197	3.382741904005422e-05	5.776047962016052e-172	bounds_lower_p=0.06268; b...	1	0	0		
pairwise_product	pairwise::producer_prozor...	Молоко питне	Молоко питне	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	reverse	ProducerUA	FarmGateUA	producerua_to_farmgateua	ARDL	FarmGateUA	ProducerUA	216	-0.03408379753109969	-0.01261089944795175	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.6093788556319114	5.311376301329217e-06	1.236267981742255e-64	bounds_lower_p=3.004e-05;...	1	0	0		
pairwise_product	pairwise::producer_prozor...	Молоко питне	Молоко питне	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	reverse	ProZorro	FarmGateUA	prozorro_to_farmgateua	ARDL	FarmGateUA	ProZorro	216	-0.000669337193433428	-0.0363319317506019	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.4008265501069578	1.468063148942169e-05	6.564381791113238e-68	bounds_lower_p=2.813e-06;...	1	0	0		
pairwise_product	pairwise::producer_prozor...	Сир твердий	Сир твердий	hard_cheese	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	forward	FarmGateUA	ProZorro	farmgateua_to_prozorro	ARDL	ProZorro	FarmGateUA	189	-8.72842418752818	-0.6419958800672612	nan	nan	nan	nan	nan	ok	0	nan	0	0	0.4763073490339259	0.7508803494476779	0.0001766670872387695	bounds_lower_p=8.513e-13;...	1	0	0		
product	product::Вершки::Silpo::o...	Вершки	Вершки	cream	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	ProZorro	Retail	prozorro_to_retail	ARDL	Retail	ProZorro	41	0.03815343029021434	-0.07780863784892769	nan	nan	nan	nan	nan	ok	0	nan	0	0	0.4763073490339259	0.7508803494476779	0.0001766670872387695	bounds_lower_p=8.513e-13;...	1	0	0		
product	product::Вершки::Silpo::o...	Вершки	Вершки	cream	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	ProZorro	farmgateua_to_prozorro	ARDL	ProZorro	FarmGateUA	41	27.40581164655441	-1.563367219701413	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.947064608010218	0.1379433157208637	0.0	bounds_lower_p=0.5287; bo...	1	0	0		
product	product::Молоко питне::Si...	Молоко питне	Молоко питне	milk	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	reverse	Retail	ProZorro	retail_to_prozorro	ARDL	ProZorro	Retail	41	-1.569298877962051	1.462494024258335	nan	nan	nan	nan	ok	0	nan	0	0	0.8435337514070991	0.9672870409827264	1.494891740049852e-10	bounds_lower_p=0.0002924;...	1	0	0			
product	product::Молоко питне::Si...	Молоко питне	Молоко питне	milk	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	reverse	ProZorro	ProducerUA	prozorro_to_producerua	ARDL	ProducerUA	ProZorro	41	-0.003981249234879858	0.08556001807837454	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.0196200671200218	0.2375437563121033	0.789219437359504	bounds_lower_p=0; bounds...	1	0	0		
product	product::Молоко питне::Si...	Молоко питне	Молоко питне	milk	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	reverse	Retail	ProducerUA	retail_to_producerua	ARDL	ProducerUA	Retail	41	-0.01904566173387146	0.1126183560663258	nan	nan	nan	nan	nan	ok	0	nan	0	0	0.3474130360123434	0.925869723100385	3.698771932262856e-05	bounds_lower_p=2.97e-07;...	1	0	0		
product	product::Сир твердий::Sil...	Сир твердий	Сир твердий	hard_cheese	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	ProducerUA	ProZorro	producerua_to_prozorro	ARDL	ProZorro	ProducerUA	38	2.533791668492441	0.8515901678330565	nan	nan	nan	nan	nan	ok	0	nan	0	0	0.8950671950008878	0.7500372663994986	0.02058974429894947	bounds_lower_p=6.579e-08;...	1	0	0		
product	product::Сир твердий::Sil...	Сир твердий	Сир твердий	hard_cheese	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	ProZorro	Retail	prozorro_to_retail	ARDL	Retail	ProZorro	38	-0.008082951427838936	0.1583981708763298	nan	nan	nan	nan	nan	ok	0	nan	0	0	0.7406617844250787	0.6046290618248675	0.8118949225796651	bounds_lower_p=0.9132; bo...	1	0	0		
product	product::Сир твердий::Sil...	Сир твердий	Сир твердий	hard_cheese	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	forward	FarmGateUA	Retail	farmgateua_to_retail	ARDL	Retail	FarmGateUA	38	-16.12941410918768	-3.273094954784757	nan	nan	nan	nan	nan	ok	0	nan	0	0	0.8090316794651169	0.5742982621103152	0.05595308976443426	bounds_lower_p=0.7505; bo...	1	0	0		
product	product::Сир твердий::Sil...	Сир твердий	Сир твердий	hard_cheese	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	reverse	ProZorro	ProducerUA	prozorro_to_producerua	ARDL	ProducerUA	ProZorro	38	0.05875093381259333	-0.0758952357905691	nan	nan	nan	nan	nan	ok	0	nan	0	0	0.8636489987653581	0.4300526930049906	0.1580768609460545	bounds_lower_p=0.0334; b...	1	0	0		
product	product::Сир твердий::Sil...	Сир твердий	Сир твердий	hard_cheese	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	reverse	Retail	FarmGateUA	retail_to_farmgateua	ARDL	FarmGateUA	Retail	38	-0.004884939285385581	-0.08156888394341669	nan	nan	nan	nan	nan	ok	0	nan	0	0	0.8807849703806206964	0.1521775431549344	0.3329413806206964	bounds_lower_p=6.961e-08;...	1	0	0		
product	product::Сметана::Silpo::...	Сметана	Сметана	sour_cream	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	reverse	ProZorro	FarmGateUA	prozorro_to_farmgateua	ARDL	FarmGateUA	ProZorro	38	-0.0002561523368839191	-0.007181451310785285	nan	nan	nan	nan	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.7838005674024638	0.002032162462147065	7.447112047328443e-08	bounds_lower_p=0.004736;...	1	0	0		
product	product::Сметана::Silpo::...	Сметана	Сметана	sour_cream	Silpo	nan	observed	linear	initial	daily	chain_common_support	full_chain	reverse	Retail	FarmGateUA	retail_to_farmgateua	ARDL	FarmGateUA	Retail	38	0.001083444100194342	-0.1056337246281378	nan	nan	nan	nan	nan	ok	0	nan	0	0	0.7373514597990629	0.6232340801443153	0.9393878293158436	bounds_lower_p=0.001988;...	1	0	0		
pairwise_product	pairwise::farm_gate_retai...	Молоко питне	Молоко питне	milk	Silpo	nan	observed	linear	initial	daily	pairwise_overlap	farm_gate_retail	reverse	Retail	FarmGateUA	retail_to_farmgateua	ARDL	FarmGateUA	Retail	48	0.0008319129231484368	-0.09123218195246746	nan	nan	nan	nan	nan	ok	0	nan	0	0	0.7373514597990629	0.6232340801443153	0.9393878293158436	bounds_lower_p=0.001988;...	1	0	0		
pairwise_product	pairwise::farm_gate_retai...	Сир кисломолочный	Сир кисломолочный	cottage_cheese	Silpo	nan	observed	linear	initial	daily	pairwise_overlap	farm_gate_retail	reverse	Retail	FarmGateUA	retail_to_farmgateua	ARDL	FarmGateUA	Retail	48	9.999156834972212e-05	-0.07799285615481645	nan	nan	nan	nan	nan	ok	0	nan	0	0	0.7373514597990629	0.6232340801443153	0.9393878293158436	bounds_lower_p=0.001988;...	1	0	0		
pairwise_product	pairwise::farm_gate_retai...	Сметана	Сметана	sour_cream	Silpo	nan	observed	linear	initial	daily	pairwise_overlap	farm_gate_retail	forward	FarmGateUA	Retail	farmgateua_to_retail	ARDL	Retail	FarmGateUA	48	-12.95339450745026	-1.906426333887981	nan	nan	nan	nan	nan	ok	0	nan	0	0	0.9724733520776613	0.1406494842016544	0.42257						